

PHASE III: Forensic Analysis Report

1. Evidence Summary

The evidence analyzed in this phase consists of a compressed ZIP file that was identified as suspicious during Phase 2. The file contained a directory with multiple items, including text files, image files, log files, and a hidden file.

- **Evidence ID:** ZIP-001
- **Evidence Type:** Compressed archive (ZIP file)
- **Source:** Local Kali Linux virtual machine
- **Acquisition Date/Time:** (Insert your actual time)
- **Chain of Custody:** Maintained within a controlled lab environment

The ZIP file was extracted, and its contents were preserved for further forensic analysis.

2. Tools and Methodology

The following tools were used to analyze the extracted files:

- **unzip** – used to extract contents of the archive
- **ls -la** – used to identify all files, including hidden files
- **cat** – used to read file contents

- **strings (optional)** – used to extract readable text from files

These tools were selected because they provide a quick and effective way to analyze file contents and identify suspicious data within a controlled environment.

The methodology followed a basic forensic workflow:

1. Identify and extract the evidence
2. Examine directory structure
3. Identify hidden or unusual files
4. Analyze file contents
5. Document findings

3. Findings and Interpretation

3.1 File Structure Analysis

After extracting the ZIP file, the following files were identified:

- notes.txt
- creds.txt
- image1.jpg
- log_file.log
- .secret.txt (hidden file)

The presence of a hidden file suggests that there may be an attempt to conceal information.

3.2 Hidden File Discovery

Using the command:

```
ls -la evidence_case
```

The file .secret.txt was discovered. Hidden files are often used to store sensitive or concealed data, making them important in forensic investigations.

3.3 Sensitive Data Identification

The file creds.txt was examined using:

```
cat evidence_case/creds.txt
```

The contents revealed:

```
password=admin123
```

This indicates that sensitive credential information was stored in plain text, which is a major security concern. In a real-world scenario, this could lead to unauthorized access.

3.4 File Content Review

Other files, such as notes.txt and log_file.log, were reviewed and did not immediately show malicious content. However, they were included as part of the investigation to ensure a complete analysis.

4. Timeline Reconstruction

Although this was a simulated environment, a basic timeline can still be inferred:

- Creation of evidence files
- Compression into ZIP archive
- Extraction and analysis during Phase 2

In a real-world case, timestamps from file metadata and logs would be used to build a more detailed timeline.

5. Conclusion and Assessment

Based on the analysis, the ZIP file contained a mix of normal and potentially sensitive data. The discovery of a hidden file and plain text credentials suggests that the archive may have been used to store or transfer information in a way that avoids detection.

While no active malware was identified in this simulation, the presence of unsecured credentials highlights a potential security risk. In a real investigation, this would require further analysis to determine how the data was created, accessed, and used.

This phase helped reinforce the importance of examining all files, including hidden ones, and carefully reviewing file contents for sensitive or suspicious information. It also prepares the investigation for the final phase, where all findings will be combined into a complete forensic case report.

6. GitHub Update

The Phase 3 report will be added to the existing repository under a new folder labeled "Phase 3." The changes will be committed with clear messages to maintain version history and document the progress of the investigation.